

ReturnGuard · architecture

Autonomous post-purchase agent that spots orders likely to be returned and steps in first · Composable AI Hackathon 2026, Track 5

1 SIGNAL

Shopify · commerce

Orders from the last 30 days (Admin GraphQL)
Line items: SKU, price, quantity
Fulfilment and return status
Client-credentials token, auto-refreshed

Databricks · Lakehouse

product_return_stats: return rate per SKU
(from real refunds in the hackathon data)
customer_return_profile: value, churn,
own return rate, support cases, consent

Bloomreach · engagement

Events since the order: sessions,
refund-policy and exchange page visits,
email opens and clicks
Turned into 7 signals (engagement.py)

Trigger: hourly schedule

No human in the loop · humans review the log

orders

features

behaviour

2 REASON (agent runtime)

ReturnGuard agent

Databricks Job · serverless · runs hourly on its own · settings from a secret scope

1 Resolve outcomes: was each earlier decision's item kept or returned?

2 Detect new orders still inside the 30-day return window

3 Assemble one context from Shopify + Databricks + Bloomreach (no PII)

4 Ask Gemini for a structured decision

5 Enforce guardrails in code (the model never has the last word)

6 Log the decision to Databricks first, then act

Google Gemini 3.8 Flash

JSON-schema-constrained output:
risk level + score
intervention and channel
message for this customer
and product, with rationale

Policy guardrails (code)

No marketing consent: no contact
Return window closed: no contact
SKU not in order: no contact
Incentive only if earned,
capped 5 / 10 / 15% by value

4 LEARN · Databricks intervention_log

Decisions + outcomes (kept / returned), read back next run: failed interventions aren't repeated

risk + event

discount

3 ACT

Bloomreach · activation

Profile: returnguard_risk_level / score
Event: returnguard_intervention
Scenario 'ReturnGuard delivery':
consent check, then email (Mailgun)
Outcome event: returnguard_outcome
Opens and clicks flow back as signals

Customer

One tailored message, or nothing:
fit guidance · usage tips
exchange offer · proactive support
small keep-incentive (high value only)

Shopify · incentive

Single-use discount code for the customer,
only when earned (sent in the message)

email

— signal in
— action out
- - learning loop

Simulated for the demo:
Shopify test orders,
storefront events (tagged)